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Numerical simulations of vented hydrogen deflagration in a medium-scale enclosure

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ABSTRACT

A validation study was performed to investigate the ability of Computational Fluid Dynamics (CFD) models to predict hydrogen deflagrations in vented enclosures. The validation exercise was aimed at assessing the suitability of CFD as a reliable tool for explosion safety assessments and involved comparing CFD predictions with measurements from an experiment carried out by FM Global in a 64 m^3 enclosure. The enclosure included a large square vent located in the center of one of its walls. The enclosure was filled with a homogenous hydrogen-air mixture of 18% v/v composition before ignition at its center. In this paper, CFD model predictions of the transient pressure and the flame speed are compared against experimental measurements. Additionally, peak overpressure predictions are compared against empirical correlations and the NFPA 68 vent sizing standard. The study focuses on the prediction of the first overpressure peak that is generated by external explosion. The agreement between the models' predictions and experimental results is found to be satisfactory, which suggests that CFD models have the potential to predict explosion phenomena with reasonable accuracy. However, more extensive model validation and sensitivity studies are required before CFD models can be used with confidence in explosion safety assessments.

1. Introduction

Flammable gases, such as natural gas and hydrogen, are used in process industries and indoor environments. In the event of an accidental release, the released gas could mix with air and form a flammable cloud. An accidental release in an enclosed space could have catastrophic consequences in the case of an explosion since confinement can lead to much higher overpressures than would be the case for an equivalent explosion in the open atmosphere. Venting of hot combustion products from an enclosure is a final protection measure used to mitigate potential damage resulting from the explosion. Although there are guidelines for the minimum vent size required for a given space (e.g. NFPA 68), these are typically very conservative (Razus and Krause, 2001) and can have limited applicability (Sustek and Janovsky, 2013). With the continual increase in computing power and resource, Computational Fluid Dynamics (CFD) modeling is becoming an increasingly attractive tool for use in risk assessment. This paper aims to assess the performance of a number of CFD models for simulating the scenario of a vented hydrogen deflagration.

Experiments conducted by Cooper et al. (1986) were used to identify the physical mechanisms leading to overpressure peaks during vented explosions. In the case of central ignition, four pressure peaks and their associated causes were highlighted: the first peak resulted from the opening of the vent; the second was caused by the external explosion; the third by the reduction of the combustion rate as the flame reached the enclosure walls; and the final peak developed due to flow instabilities enhancing combustion through flame-acoustic interactions. The magnitude of these pressure peaks depends on a number of factors, including the geometry of the confined space, the vent size, the mixture composition and the ignition location.

In the case of central ignition (or other cases where ignition is not close to a vent), as the flame propagates away from the ignition point, unburned mixture is forced out through the vent. If the flame also passes through the vent, any accumulated flammable mixture outside the enclosure can ignite, initiating an external explosion. Once the external explosion has been initiated, it occurs concurrently with the

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Nomenclature

S_u	Burning velocity (m/s)
S_u^w	Subgrid burning velocity (m/s)
S_L	Laminar burning velocity (m/s)
S_{QL}	Quasi-laminar burning velocity (m/s)
S_t	Turbulent burning velocity (m/s)
u'	SGS or fluctuating velocity component (m/s)
l_t	Integral turbulence length scale (m)
l_I	Characteristic length scale (m)
λ_u	Thermal diffusivity of the unburned mixture (m^2/s)
G	Stretching factor
c	Progress variable
R	Flame radius (m)
R_0	Critical radius (m)

Greek letters

ρ	Mixture density (kg/m^3)
μ	Dynamic viscosity (kg/m/s)
Ξ_k	Wrinkling factor due to turbulence generated by the flame front itself
Ξ_{lp}	Wrinkling factor due to leading point mechanism
Ξ_f	Wrinkling factor due to fractal increase of flame surface area
Ξ_{RT}	Wrinkling factor due to Rayleigh-Taylor instability
ψ	Model parameter
E	Expansion ratio
ε	Thermokinetic index

Subscripts, superscripts and bars

0	Initial conditions
u	Unburned mixture
t	Turbulent
max	Maximum

ongoing internal combustion.

The interaction between the internal and external explosion is of critical importance as a number of studies have highlighted a correspondence between the onset of the external explosion and the timing of the peak enclosure pressure. Harrison and Eyre (1987) observed external explosions in their vented deflagration tests and noted a possible influence on the magnitude of the internal overpressures. Jiang et al. (2005) performed several vented deflagration experiments, studying the effect of vent cover failure pressure, ignition location, vent area and fuel concentration, in order to investigate the characteristics of external explosions. Proust and Leprette (2009) highlighted the significance of the external explosion by performing a series of tests using various vessel sizes and fuels. Chao et al. (2011) carried out tests to analyze peak overpressures in vented explosions and examined the influence of fuel, enclosure size, ignition location, vent size and obstacles. The results indicate that the magnitude of the internal overpressure can depend on the characteristics of the external explosion.

According to Cooper et al. (1986) the internal overpressure peak following the onset of external explosion often has the largest magnitude (out of the four peaks identified). One explanation for this behavior is the highly energetic nature of the external explosion due to high levels of turbulence in the unburnt gas outside the vent (resulting from the jet-like flow of unburned gas through the opening) and the manner in which the external explosion is initiated (by a highly turbulent jet flame). An alternative explanation, which has been proposed by Tsuruda and Hirano (1987), highlights the Rayleigh-Taylor (RT) instability, which develops when gas of low density is accelerated into a region of higher density gas, as a dominant mechanism for turbulence generation in deflagrations. The influence of the RT instability has been incorporated into two of the models used in the model validation exercise presented here.

CFD modeling of the coupled flow and combustion phenomena in vented deflagrations presents a significant computational challenge. Combustion occurs at a scale which is several orders of magnitude smaller than the scale typical for geometries of practical interest. Turbulence plays a significant role in combustion acceleration and deflagration dynamics while modeling turbulence in reacting flows remains a demanding task (one may note that even the modeling of isothermal turbulent flow is a subject of open research). The growth of instabilities at the flame front, the complex interaction between the flame and the turbulent flow along with the very wide range of applications make deflagration modeling a complex task. Classical CFD models cannot predict vented deflagrations with ignition near the vent (front ignition), as for example it is shown in Bauwens et al. (2011), because of the complex interactions between the flame front and the acoustic waves generated from to the structural response of the enclosure. Combustion modelling of lean hydrogen mixtures has an additional challenge compared to other cases because of the strong non-equidiffusion effects which increase complexity and affects the flame-turbulence interaction (Lipatnikov and Chomiak, 2005).

Baraldi et al. (2010) performed a comparison of CFD models for hydrogen deflagrations in a 0.95 m^3 vented vessel and found that the maximum overpressure was predicted with reasonable accuracy. They identified a need for blind prediction studies as a means of testing model performance against experimental data more rigorously. Bauwens et al. (2011) and Keenan et al. (2014) simulated hydrogen deflagrations in a 64 m^3 vented enclosure with a focus on modeling the RT instability. The instability was modeled using an additional transport equation to account for the rates of generation and suppression of the instability. Large Eddy Simulation (LES) studies of small-scale vented deflagrations have been conducted by Di Sarli et al. (2009) and Quillatre et al. (2013), which showed very good predictive capabilities of their respective models. The importance of accurate external explosion modeling was highlighted in the work of Molkov et al. (2006) who performed LES modeling of large-scale (547 m^3) deflagration tests. Simulation of explosions vented through relief pipes have also been performed by Ferrara et al. (2006) and Makarov et al. (2007) with success.

The primary objective of this work is to assess the performance of a number of CFD models for simulating vented hydrogen deflagrations at the scales typical for practical applications by performing simulations of a vented hydrogen deflagration experiment conducted by FM Global. This study considers hydrogen deflagrations as a consequence of increased hydrogen use in industry and the safety issues resulting from its explosive nature. Model performance for predicting internal and external overpressure, flame and flow velocity, hydrogen concentration, turbulent fluctuations and flame speed is assessed. Sensitivity analyses of several parameters were conducted in order to examine their impact on the results, following Best Practice Guidelines (BPG) such as ERCOFTAC (2000) and the recently developed BPG for hydrogen applications (Baraldi et al., 2016; Tolias et al., 2017). The effect of external explosion is investigated and discussed. The results of empirical

correlations and the NFPA 68 standard are also presented and compared against both the experimental and CFD results.

2. FM Global experiment

A series of vented deflagration experiments with various flammable gases, vent openings and ignition locations were performed by FM Global and have been described by Bauwens et al. (2008, 2009, 2011). Deflagration experiments using a uniform 18% v/v hydrogen-air mixture composition with central ignition and a 5.4 m^2 square vent are chosen for this validation exercise. The experimental facility consists of a square-floored enclosure of 4.6 × 4.6 × 3.0 m (length x width x height) with a vent approximately 2.32 × 2.32 m in size located in the center of one of the walls (Fig. 1). A porch, consisting of steel beams and a concrete slab, extended perpendicularly from the vent, as shown in Fig. 3.

The exact dimensions of the beams and the concrete slab outside the enclosure are not known. However, they are not expected to have a significant impact on the results. The following dimensions (extracted from Fig. 2 of Bauwens et al., 2011) can be applied approximately: Concrete slab dimension 4.8 × 2.0 × 0.3 m, beam dimension 0.2 × 0.2 × 2.6 m, beam distances from the vent 0.0, 0.8, 2.1, 3.4 and 4.7 m.

Four pressure transducers were mounted in the chamber. No significant differences were found between the pressure measurements from the sensors during the experiment (Bauwens et al., 2011). The record of pressure dynamics from sensor P1 (located at the center of the right wall, see Fig. 1) is provided by Bauwens et al. (2011) to compare model predictions against.

Twenty flame time-of-arrival thermocouples were placed inside and outside the enclosure (Bauwens et al., 2011). A record of flame propagation velocity for deflagration of a 17.8% v/v hydrogen-air composition is available from Bauwens et al. (2008); the record of flame velocity for deflagration of an 18.2% v/v hydrogen-air mixture is given by Bauwens et al. (2009), see Fig. 2 below. The two curves are very similar, indicating a very high level of experimental repeatability. Note that in Fig. 2 the horizontal distance is measured from the ignition location and, thus, the vent is located at 2.3 m. Finally, fans were installed to stir the hydrogen-air mixture and provide an initial turbulence velocity field; turbulent velocity fluctuations $u' = 0.1\ m/s$ were measured immediately prior to ignition.

3. Modeling approach

The FM Global experiment was simulated by the three organizations: Health and Safety Executive, UK (HSE), Ulster University, UK (UU), and National Centre for Scientific Research Demokritos, Greece (NCSRD), using four different CFD codes. HSE used GexCon FLACS v10.2 and ANSYS CFX v16.0, UU used its own combustion model realized based on ANSYS FLUENT v14.5 platform, and NCSRD used ADREA-HF. This section describes the modeling approach for each code. Additional numerical details are presented in Tables 1–3.

3.1. FLACS simulations

FLACS is a widely used software package, primarily for explosion and dispersion modeling. The FLACS simulations presented here were carried out following guidelines and advice given by the model developer, GexCon. The results are, therefore, representative of those a typical, trained user should obtain.

FLACS is set up to enable users to quickly and simply define and run gas explosion scenarios with minimal input. It uses a Porosity Distributed Resistance (PDR) model to account for the presence of blockages to the flow. The equations solved are the conservation of mass, momentum, enthalpy, fuel mass fraction and mixture fraction, which are defined in the FLACS user manual (FLACS, 2015) for the PDR method. A RANS (Reynolds-Averaged Navier Stokes) approach to turbulence modeling is incorporated with the standard $k - \varepsilon$ model.

Flame modeling is handled by a beta-flame model (or β -model) in which the flame front is augmented by multiplying the diffusion coefficient for the fuel and dividing the reaction rate by a common factor β. The value of β is proportional to the grid size (Arntzen, 1998), therefore truly grid independent solutions cannot be obtained. Three parameters for the calculation of the burning velocity are used. First the burning velocity is set equal to the laminar burning velocity, S_L, of the gas mixture, determined from a correlation. Then, the burning velocity is calculated based on the quasi-laminar burning velocity S_{QL} and the turbulent burning velocity S_t (see Table 3).

Ignition is modelled by artificially initiating the combustion reaction at the start of a simulation in the cell in which the ignition location is specified (Middha, 2010). The local turbulence conditions and the grid cell size, both of which influence flame propagation, have been specified as per the FM-Global turbulence measurements and the FLACS user guidelines (FLACS, 2015), respectively.

To account for the high reactivity of hydrogen and to incorporate the effects of increased flame wrinkling and flame acceleration for lean hydrogen-air mixtures, FLACS contains a number of hydrogen-specific combustion modifications. For lean hydrogen-air mixtures, where the Lewis number is less than unity, the flame temperature is increased so as to lead to an increase in flame speed. The effect of the Lewis number on flame wrinkling is also incorporated through an increase in laminar flame speeds for lean hydrogen-air mixtures. This results in increased wrinkling at the flame front and more substantial flame acceleration for lean mixtures as compared to mixtures which are hydrogen-rich (Middha, 2010).

The geometry used for the simulations was constructed to match the experimental setup used by FM Global. A domain size of 20.0 m × 14.4 m x 12.0 m was used for the simulations, corresponding to roughly twice the module (enclosure plus porch) length by three times the enclosure width by four times the enclosure height. Grid resolutions of 30, 20, 10, 5 and 2.5 cm were investigated. The grid was refined to the above resolutions within a bounding box volume containing all of the geometry (including the porch); the grid was then expanded to the domain boundaries. This gives cubic cells in the region of interest, which is necessary within FLACS for the prediction of gas explosions, and limits the grid requirement in regions where the results are less important.

The time-step is selected based upon two Courant numbers: the first, CFLC, is based on the speed of sound; the second, CFLV, is determined using local flow velocities. The first of these is used to limit the extent to which the pressure (sound) wave can travel during a single step of the simulation. The second limits the propagation of fluid velocity during a single time step. The most stringent time-step value is chosen from the two CFL numbers. The suggested values of 5 and 0.5 were used respectively.

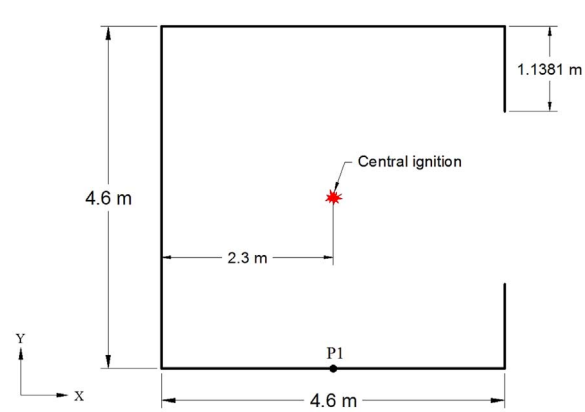

Fig. 1. Main dimensions of the experimental set up.

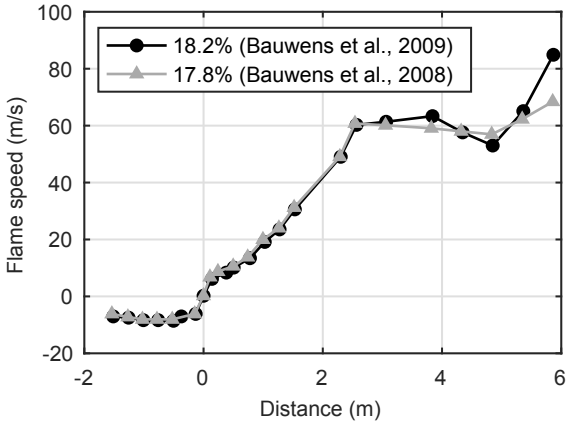

Fig. 2. Experimental flame speed.

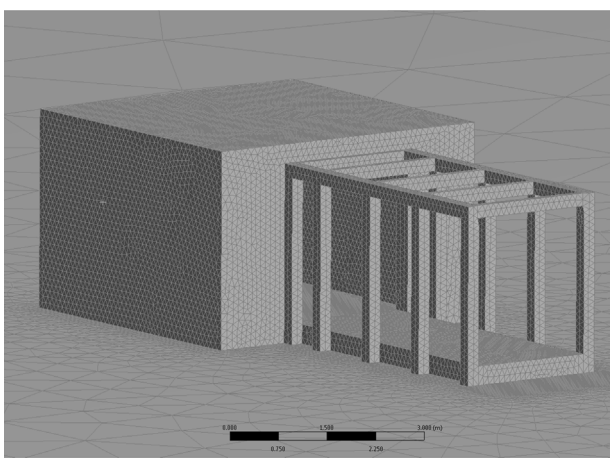

Fig. 3. Model geometry used for simulations using CFX.

The initial turbulent fluctuation velocity was specified for the simulations as $u' = 0.1$ m/s, corresponding to the condition observed during the FM Global experiment. The initial turbulence length scale, used to define the initial rate of turbulent dissipation, was specified at 10% of the minimum grid cell size for each simulation; this is consistent with the FLACS user guidelines. At the domain boundaries the FLACS *Euler*-type boundary condition was used, where the pressure was fixed to ambient pressure.

3.2. CFX simulations

CFX is a general purpose CFD software package capable of simulating a wide range of fluid flows. There is a wide choice of turbulence and combustion models. For the simulations described here, turbulence was modeled using the k-ε model with buoyancy production and dissipation. Heat transfer was modeled using the total energy model and combustion was modeled using the Burning Velocity Model (BVM) with laminar flamelets (ANSYS, 2015).

The BVM model requires a sub-model for the turbulent flame speed closure (TFSC) and the Zimont TFSC model, which is available as standard in CFX (ANSYS, 2015) was used. This model has previously been used to model hydrogen deflagrations by Bender and Menter (2002) who used it to simulate flame acceleration in the RUT facility, Hoyes et al. (2013) who used it to model slow isochoric deflagrations in the THAI facility and Sathiah et al. (2014) who used it to model fast deflagrations in the ENACCEF facility.

The model geometry as represented in the CFX model is shown in Fig. 3. The dimensions of the enclosure, vent and porch were chosen to match the experimental set-up. The enclosure was surrounded by an external region consisting of a 100 m cube. Simulations were carried out on tetrahedral meshes with mesh refinement used near the ignition location, inside the enclosure, outside the vent and inside the porch for all mesh resolutions considered. Various computational grids were investigated. The grid resolution inside the enclosure for each mesh was equal to 15, 12.5, 10, 7.5 and 5 cm with refinement in the region of ignition site roughly half this size. The corresponding total number of computational nodes was equal to 0.05, 0.13, 0.22, 0.59 and 1.4 million respectively.

The sensitivity of vented hydrogen deflagration predictions to the initial level of turbulence is well documented (e.g. Bauwens and Dorofeev, 2014; Hooker et al., 2015). Zimont TFSC simulations were carried out using an initial turbulent kinetic energy of 0.015 m^2/s^2 (equivalent to turbulent velocity fluctuations of 0.1 m/s – as measured in the FM Global experiment) and an initial turbulent eddy dissipation determined automatically by CFX. However, these results were found to be highly sensitive to both the mesh resolution and the time step size. ANSYS suggested using artificially high initial levels of turbulent kinetic energy and turbulent eddy dissipation and, therefore, the Zimont-TFSC simulations described here were carried out using an initial turbulent kinetic energy of 1 m^2/s^2 and an initial turbulent eddy dissipation of 10 m^2/s^3. These values were used by Bender and Menter (2002) to model lean hydrogen deflagrations.

In all of the simulations the initial hydrogen concentration was set to match the experimental conditions of 18% v/v inside the enclosure. The initial velocity was set to zero everywhere, whilst the initial temperature was set to 12 °C. The walls of the enclosure and porch were modeled as no-slip, adiabatic walls. The ground was also modeled as a no-slip, adiabatic wall whilst the top boundary of the external region was modeled as a pressure controlled opening.

Ignition was achieved using the inbuilt model and sensitivity to the ignition parameters was not investigated. Sensitivity tests were performed on the solver parameters to determine the most appropriate time step size and the required number of sub-iterations. At least 5 sub-iterations were needed per time step and the time step size was set to

Table 1
Main numerical parameters used in each code.

Code	Solver	Discretization scheme C = convective terms T = temporal terms	Time step requirements
FLACS	Compressible extension to the SIMPLE pressure-correction algorithm	C = 2nd order hybrid scheme (weighting between 2nd order upwind and 2nd order central differencing schemes) (Middha and Hansen, 2009) T = 1st order backward Euler (implicit)	CFLV = 0.5 and CFLC = 5
CFX	Compressible based on SIMPLE algorithm	C = high resolution advection scheme T = 2nd order backward Euler (implicit)	1.0 ms
FLUENT	Coupled explicit compressible solver	C = 2nd order upwind T = 4-stage Runge-Kutta scheme (explicit)	CFL < 0.8
ADREA-HF	Modified SIMPLER	C = bounded central scheme for momentum equations, bounded linear upwind scheme for species and energy equations T = Crank Nicolson	CFL = 0.1

Table 2
Computational domain, computational grid and CPU time for each code.

Code	Computational Domain	Grid type	Grid resolution	CPU time and computational system
FLACS	Rectangular box of 20 × 14.4 × 12 m (L × W × H)	Structured Cartesian mesh	Δ_{CV}=0.05 m (uniform inside the enclosure and porch, then cell size is increased to the domain boundaries by an expansion factor of 1.1) CVs = 3,000,000	Time: 6 h for a 1 s physical time, parallel across 20 cores System: two Intel Xeon E5-2697 v2 (2.7 GHz) CPUs
CFX	Cubic box of 100 m	Unstructured tetrahedral	Δ_{CV} = 0.05 m (uniform inside the room) Nodes = 1,400,000	48 h for a 0.6 s physical time, parallel across 22 cores System: two Intel Xeon E5-2697 v2 (2.7 GHz) CPUs
FLUENT	Hemispherical of radius 25 m	Unstructured tetrahedral	Δ_{CV} = 0.10 m (average edge size inside the room and in the area surrounding the vent) CVs = 1,037,500	80 h (to calculate 0.45 s), parallel across 32 cores. System: four AMD Opteron 6376 (2.3 GHz) CPUs
ADREA-HF	Rectangular box of 42 × 60 × 30 m (L × W × H)	Structured Cartesian mesh	Δ_{CV}=0.05 m (approximately uniform inside the room and outside of it until 1.0 m distance from the vent, then cell' size is increased with an expansion factor of 1.08) CVs = 2,751,534	Time: 315 h for 0.45 s physical time, parallel across 4 cores System: Intel Xeon E5-2637 v2 (3.5 GHz) CPU

Table 3
Turbulence and combustion models used in each code.

Code	Turbulence model	Combustion model	Expression for burning velocity
FLACS	k-ε	β – flame model (Arntzen, 1998)	$S_u = \max(S_{QL}, S_t)$ $S_{QL} = S_L\left(1 + c\cdot\min\left(\left(\frac{R}{3}\right)^{0.5}, 1\right)\right)$ $S_t = 1.81u'^{0.412}l_I^{0.196}S_L^{0.784}\mu^{-0.196}$
CFX	k-ε	Zimont model	$S_t = AGu'^{3/4}S_L^{1/2}\lambda_u^{-1/4}l_t^{1/4}$ $A = 0.5$
FLUENT	RNG LES	Multi-phenomena turbulent burning velocity deflagration model	$S_t = \Xi_k\cdot\Xi_{lp}\cdot\Xi_f\cdot\Xi_{RT}\cdot S_L\cdot\exp\left(\frac{u'}{S_t}\right)^2$ Parameters: S_{L0} = 0.64 m/s, R_0 = 1.0, $E = 5.14$, $\psi = 0.5$, $\varepsilon = 0.65$, $\Xi_{lp}^{max} = 2.1$
ADREA-HF	RNG LES	Multi-phenomena turbulent burning velocity deflagration model	$S_t = \Xi_k\cdot\Xi_{lp}\cdot\Xi_f\cdot\Xi_{RT}\cdot S_L\cdot\exp\left(\frac{u'}{S_t}\right)^2$ Parameters: S_{L0} = 0.64 m/s, R_0 = 1.0, $E = 5.2$, $\psi = 0.90$, $\varepsilon = 0.64$, $\Xi_{lp} = \Xi_{lp}^{max} = 1.96$, $\Xi_{RT} = 2.3$

1 ms. Numerical convergence of the simulations was assessed by following guidelines in the CFX User Guide (ANSYS, 2015). Calculations were judged to be numerically converged when the RMS residuals were below 10^{-4}.

3.3. FLUENT simulations

The multiphenomena deflagration model relying on LES approach for turbulence simulation has being developed at Ulster University during the last decade and realized using the ANSYS FLUENT platform as a CFD engine. It is based on the turbulent burning velocity combustion sub-model taking into account physical phenomena responsible for flame wrinkling and combustion acceleration at large scales. The detailed model description and references to validation studies may be found elsewhere, e.g. Molkov (2012), and only the main model features are given below.

The governing equations were obtained by filtering three dimensional conservation equations of mass, momentum, energy and species concentration. Due to the large scale of experimental problem simulated in this validation exercise, the effects of turbulence and flame instabilities on combustion acceleration cannot be resolved and should be modeled.

The progress variable equation to simulate flame propagation and the gradient method for mass burning rate are at the core of the combustion model. The progress variable is defined as the mass fraction of combustion products. The source term in the progress variable equation describes mass burning rate and can be written using the gradient method (Prudnikov, 1967) as:

$$\bar{S}_c = \rho_u S_t |grad\ \tilde{c}|, \tag{1}$$

where $|grad\ \tilde{c}|$ is the magnitude of the gradient of the progress variable. The gradient method returns the value of the mass burning rate per unit area of the resolved flame front as specified by the term ($\rho_u\ S_t$). The method is independent of the mesh resolution, which reduces combustion modeling to the prediction of the turbulent burning velocity S_t. The effective viscosity, effective Prandtl and Schmidt numbers are calculated according to the renormalization group (RNG) theory (Yakhot and Orszag, 1986).

The model accounts for the following phenomena, which lead to flame wrinkling and combustion acceleration: turbulence generated by the flame front itself (described using flame wrinkling factor Ξ_k),

preferential diffusion instabilities at the flame front, accounted for via the leading point concept (flame wrinkling factor Ξ_{lp}), fractal nature of flame surface development (wrinkling factor Ξ_f) and flame wrinkling due to RT instabilities (factor Ξ_{RT}). The equation describing the turbulent burning velocity is presented in Table 3. This equation is a modified form of Yakhot's original equation (Yakhot, 1988). The key step in the development of this model is the substitution of the laminar burning velocity term in Yakhot's original equation with the sub-grid scale (SGS) wrinkled burning velocity S_u^w, where

$$S_u^w = \Xi_k \cdot \Xi_{lp} \cdot \Xi_f \cdot \Xi_{RT} \cdot S_L \quad (2)$$

Here it is assumed that the phenomena described by wrinkling factors act at scales comparable to the flame front thickness and below turbulence length scales. It should be noted that S_u^w influences the total turbulent burning rate through interaction with flow turbulence in the unburned mixture. The development of flame wrinkling factors Ξ_k, Ξ_{lp}, Ξ_f was modelled as a function of flame radius R and the critical radius for the onset of fully turbulent regime of flame front propagation R_0.

The Ξ_{lp} factor is very significant in lean hydrogen-air mixtures because of the strong non-equidiffusive effects and is calculated based on the concept of the leading point mechanism (Zimont and Lipatnikov, 1995). The characteristic radius for transition to fully developed flame wrinkling (where Ξ_{lp} obtains a constant value equal to Ξ_{lp}^{max} – see Table 3) due to leading point mechanism is assumed to be equal to the half of the characteristic radius R_0. The characteristic radius R_0 at which transition to the fully turbulent self-similar regime occurs is set to 1 m (Gostintsev et al., 1989). The 'ad-hoc' parameter ψ contained within the model ($\psi < 1$) is set to 0.5 following Verbecke (2009). The dependence of the laminar burning velocity on transient pressure and temperature is taken into account following the assumption of adiabatic compression/expansion. This requires the calculation of the thermokinetic index which is taken here as $\varepsilon = 0.65$ (Babkin, 2003).

The model for flame wrinkling due to RT instability was added to the FLUENT simulations presented here to check the influence of RT instability on deflagration dynamics. Analysis of flame perturbation development and the resultant model equations are described in detail by Keenan et al. (2014). Alternative RT instability model was used for deflagration dynamics simulations in Bauwens et al. (2011). It also solves an additional transport equation for the flame surface wrinkling due to RT instability, and utilizes the same equation describing amplitude growth. However, the model described by Bauwens et al. (2011) assumes that wavelength is constant and amplitude of wrinkling is much smaller than a quarter wavelength, whereas the model described here does not rely on these strong assumptions.

The calculation domain in the FLUENT simulations comprises a large hemispherical area of radius 25 m (Fig. 4). It was meshed using an unstructured tetrahedral grid. The smallest CVs, with an average edge size of 0.1 m, were located inside the chamber and in the area surrounding the vent. The characteristic CV size was then smoothly increased within the rest of the calculation domain.

The boundary conditions applied were non-slip, non-permeable, adiabatic boundary on all walls and ground surfaces. At the outer edge the non-reflecting pressure far-field boundary conditions, as implemented by FLUENT based on Riemann invariants, were applied. The model assumed an unrestricted open vent. As initial conditions, a stagnant flow field with no turbulence was specified. The flammable mixture was initially contained inside the chamber, with hydrogen mass fraction $Y_{H_2} = 0.015$ (corresponding to 18% of hydrogen by volume) and air mass fraction $Y_a = 0.985$ ($Y_a = 1.000$ in the remaining area). Initial pressure was equal to atmospheric one, initial temperature was 295 K. The initial value of the progress variable was $c = 0$. Combustion was initiated by slowly increasing the progress variable in one CV. Duration of ignition Δt_{ign} was assumed to be equal to the flame propagation time from the center to the edge of the ignition CV: $\Delta t_{ign} = 0.5\Delta_{cv}/(E \cdot S_u)$ where Δ_{CV} is the characteristic size of the cell.

3.4. ADREA-HF simulations

NCSRD used their in-house CFD code called ADREA-HF. It is a general purpose CFD code which has been extensively validated against dispersion experiments (both gaseous (Venetsanos et al., 2010) and cryogenic (Giannissi et al., 2013, 2014) releases) and combustion experiments (Tolias et al., 2014, 2015, 2016). ADREA-HF models the transient three-dimensional turbulent flow field and the dispersion of a multispecies mixture. The mixture is assumed to be ideal and in thermodynamic equilibrium.

The space-averaged Navier-Stokes equations along with the energy equation (conservation equation of static enthalpy) and the conservation equation of each of the mass fraction of the species that take part in the combustion process are solved. The equation of state for ideal gases relates pressure and enthalpy with density and temperature. Turbulence is modeled using the RNG LES. The set of main equations used are presented in (Tolias et al., 2014).

The combustion model implemented in the ADREA-HF code (Tolias et al., 2014) is the multi-phenomena turbulent burning velocity model described in the previous section. The model parameters that were used are presented in Table 3. There are two main differences in the model as used for the ADREA-HF simulations compared to the model as implemented in FLUENT. Firstly, the maximum value of the Ξ_{lp} factor was imposed from the beginning of the explosion (no transient area). Secondly, a constant value of the Ξ_{RT} factor was chosen in order to reproduce the reaction rate enhancement due to the RT instability. Its value was set equal to 2.3 at the area immediately outside the vent where the RT instability develops and equal to 1.0 elsewhere. A similar modeling approach has been used to model a large scale vented methane explosion (Molkov et al., 2006). Although the Ξ_{RT} value is case-dependent, Tolias et al. (2015) show that the same value can be used to simulate back-wall and central ignition deflagration cases of the same experiment equally well. As a result simple relations may exist which correlate Ξ_{RT} with experimental parameters.

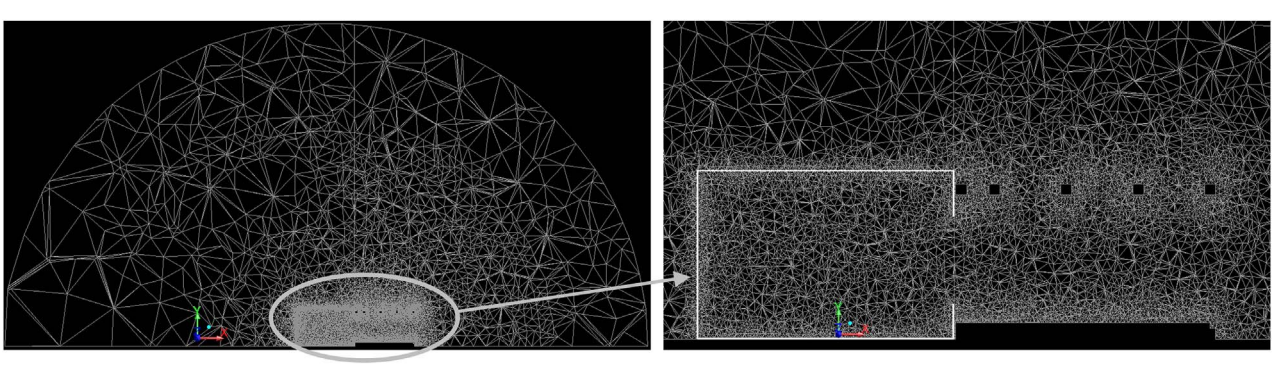

Fig. 4. Computational domain and grid used for FLUENT simulations: overall view of domain (left), enlarged enclosure view (right).

ADREA-HF uses the finite volume method on a staggered Cartesian grid. The computational domain was extended in all directions outside the room in order to minimize the effect of the imposed boundary conditions. The size of the computational cells was near uniform inside the room and at the area outside the vent. The cell volume was increased gradually in the area away from the vent in order to save computational time. Four computational grids were used in order to assess grid sensitivity. The grid resolution inside the room for each grid was equal to 10, 7.5, 5.0 and 3.5 cm and the total number of CVs in the whole computational domain equal to 0.7, 1.5, 2.8 and 6.2 million respectively. The grid of 5 cm resolution is shown in Fig. 5.

In all exit planes (lateral, front, back and top) the non-reflecting type boundary conditions for the normal velocities was chosen, while for the velocity components parallel to the exit planes, Neumann boundary conditions were applied. Zero gradient was utilized also for the mass fraction of species. As initial conditions, a stagnant flow field with no turbulence was set. In the hydrogen-air premixed area an initial mass fraction of hydrogen and oxygen was specified, equal to 0.01511 and 0.2294, respectively. Outside this area the mass fraction of oxygen was set equal to 0.232. Nitrogen was the inert specie. Ignition was modeled in the same way as in FLUENT simulation. The duration of the ignition was estimated equal to 0.0075 s for the case of 5.0 cm grid resolution. Our experience has shown that the value of this duration has weak influence on the results.

4. Results and discussion

4.1. Sensitivity analyses - FLACS

Sensitivity tests using *Euler* and *Plane-wave* pressure boundary conditions were undertaken, which resulted in a peak overpressure variation of less than 1.4%. Sensitivity to the location of the domain boundaries was also assessed with only minor differences when the domain size was extended beyond 20 m × 14.4 m x 12 m. *Euler* type boundary conditions were used for the FLACS simulations presented here. In Fig. 6, the peak overpressure at 3.5 m outside the vent is presented, for two types of boundary conditions and for different domain heights.

A grid sensitivity analysis was performed. Mesh resolutions of 30 cm, 20 cm, 10 cm, 5 cm and 2.5 cm were chosen to give grid resolutions ranging from the minimum to the maximum as advised by GexCon for gas explosion simulations at this scale. Fig. 7 shows a comparison of the predicted flame speed and pressure at the P1 sensor location for each mesh against the experimental data. The solid black lines within the figure represent the measured flame speed (left) and pressure (right). The results from the 10 cm, 5 cm and 2.5 cm meshes are similar; the solution can therefore be considered to be adequately independent of the grid over this range. The 5 cm mesh was used to produce the final FLACS results for the model validation exercise, having approximately 3 million grid cells.

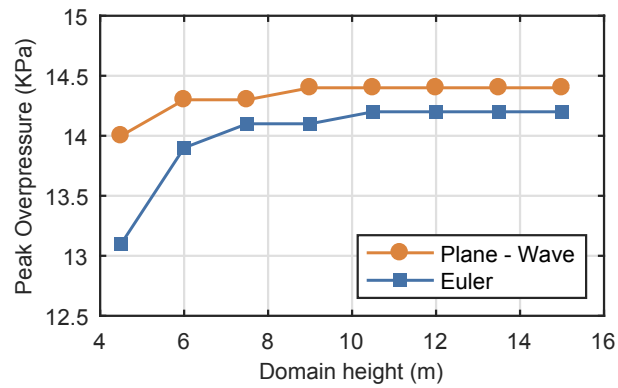

Fig. 6. Sensitivity study of boundary conditions and domain height (FLACS): Peak overpressure at 3.5 m outside the vent.

A final sensitivity analysis was performed to establish the implications of initial turbulence length scale choice using a mesh with 5 cm resolution in the region of interest. Initial turbulence length scales ranging from 1% to 20% of the grid size were used. The peak overpressure at the P1 sensor location was found to increase by more than a factor of 2.5 from the 1%–20% cases. As such, it is clear that the peak overpressure predictions from FLACS simulations are heavily dependent upon the initial turbulence field and that the initial turbulence settings could, in theory, be used to tune the result so as to give better agreement with the measured result. For the model validation presented here, an initial turbulence length scale of 0.5 cm, corresponding to 10% of the grid size, was used.

4.2. Sensitivity analyses - CFX

Sensitivity studies of the domain size, performed by varying the domain size by an order of magnitude, indicated a good level of domain size independence: only a 1% change to the timing and magnitude to the peak overpressure is attributable to the change in size. A further sensitivity study to the effect of initial turbulence conditions found, as expected, a large level of sensitivity. The values of the initial turbulence conditions were therefore chosen to be the same as Bender and Menter (2002).

Simulations were carried out to assess the sensitivity of the CFX predictions to the mesh resolution. Fig. 8 shows flame speed and overpressure predictions from simulations carried out using the five considered mesh resolutions. The flame speed predictions are largely mesh independent. Whilst the simulation on the coarsest mesh has a maximum flame speed of 72 m/s, all four simulations on the finer

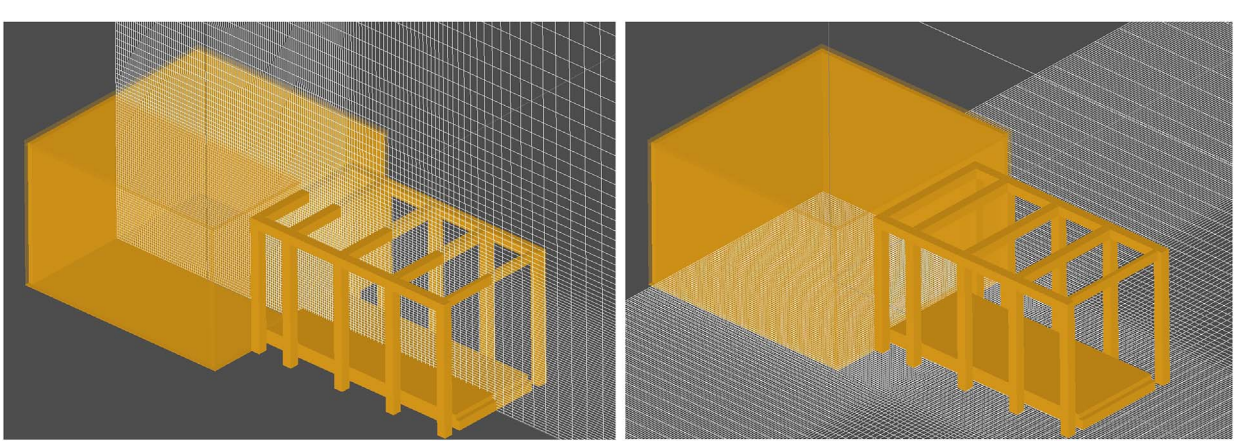

Fig. 5. Computational grid (5 cm resolution) in xz and xy planes for the ADREA-HF simulations.

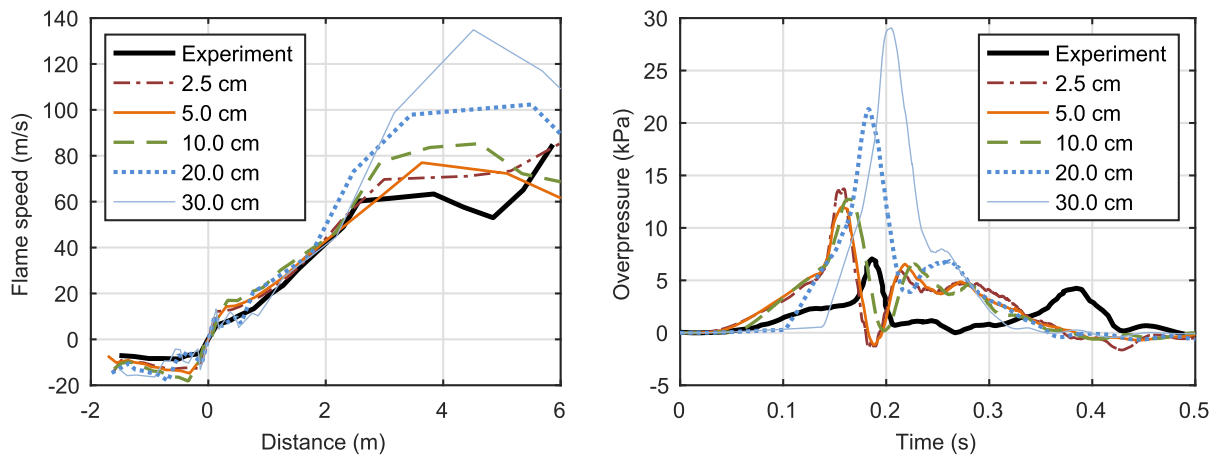

Fig. 7. Comparison of experimental and simulated (FLACS) flame speed and overpressure for multiple mesh resolutions.

resolution meshes have a maximum flame speed of about 86 m/s. The simulation on the coarsest mesh gave a peak overpressure of about 5 kPa, whilst the simulations on the 12.5 cm, 10 cm and 7.5 cm meshes gave values between 6.3 kPa and 6.7 kPa, and the simulation on the 5 cm mesh gave a prediction of 7.7 kPa. The overpressure predictions indicate only a moderate level of mesh independence. Comparison to the experimental data indicates that the simulation results are converging to a maximum flame speed and peak overpressure which are larger than the experimental values. The general trend with increasing mesh resolution is that the peak overpressure increases. The results of the 5 cm mesh are presented in Section 4.4.

4.3. Sensitivity analyses - ADREA-HF and FLUENT

Higher resolution grids were tested for ADREA-HF model to assess the sensitivity of predictions to the mesh resolution. Fig. 9 presents the comparison of the predicted flame speed and pressure at the P1 sensor location for each mesh against the experimental data. Four different meshes were used, with cell size inside the enclosure of 3.5, 5.0, 7.5 and 10.0 cm. No significant changes in the results are observed. The results of the 5 cm mesh consisting of 2,751,534 CVs are presented in Section 4.4. In FLUENT, the same model realization was tested for grid sensitivity using different deflagration cases elsewhere (Verbecke, 2009) and was found to be independent on grid resolution when major deflagration features are resolved or properly modelled using grid non-sensitive techniques for flame propagation like the progress variable equation with the gradient method for source term.

Domain size was chosen based on experience of the organizations using the codes with the chosen dimensions not expected to influence the results. A sensitivity study on the Ξ_{RT} factor was also performed using ADREA-HF. The influence of this factor on the model predictions is an increase of the first pressure peak with an increase of the factor Ξ_{RT}. This is an expected behavior because higher values of Ξ_{RT} means more violent external explosion. Simulations without the Ξ_{RT} factor were also performed by both partners. It was concluded that a reaction rate enhancement mechanism needs to be included in the model in order to simulate external explosion properly. This remark is also shown in Keenan et al. (2014) and Tolias et al. (2015).

4.4. Results

Fig. 10 presents the overpressure time history inside the enclosure (at sensor P1) for the experiment (filtered results) and the simulations. Two main overpressure peaks are observed in the experiment: the first one is 7.0 kPa at 0.19 s and the second one is 4.2 kPa at 0.38 s. The unfiltered experimental results, which are presented by Bauwens et al. (2011), exhibit high frequency oscillations after 0.2 s. According to Cooper et al. (1986) and Bauwens et al. (2011), the second overpressure peak is a result of flame-acoustic interaction and the structural response of the room. A sub-model for this complex interaction is not included in any of the models used here and consequently, the prediction of the second pressure peak is not expected.

Regarding the first overpressure peak, we observe that both ADREA-HF and FLUENT predict it very accurately. On the other hand FLACS

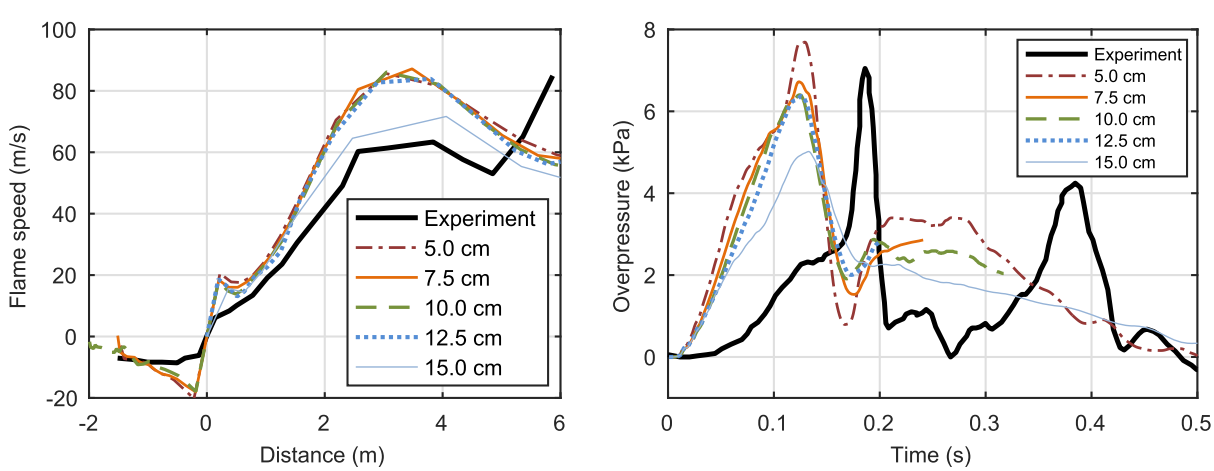

Fig. 8. Comparison of experimental and simulated (CFX) flame speed and overpressure for multiple mesh resolutions.

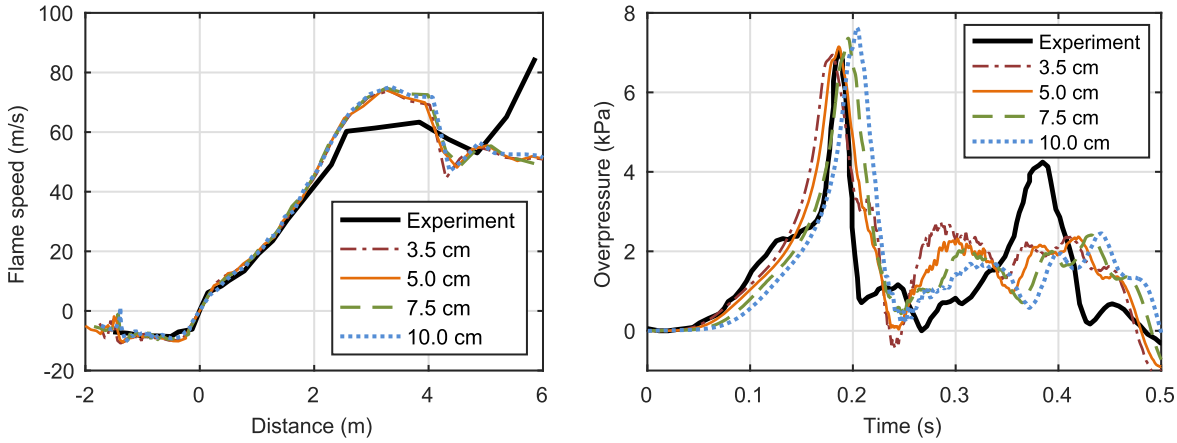

Fig. 9. Comparison of experimental and simulated (ADREA-HF) flame speed and overpressure for multiple mesh resolutions.

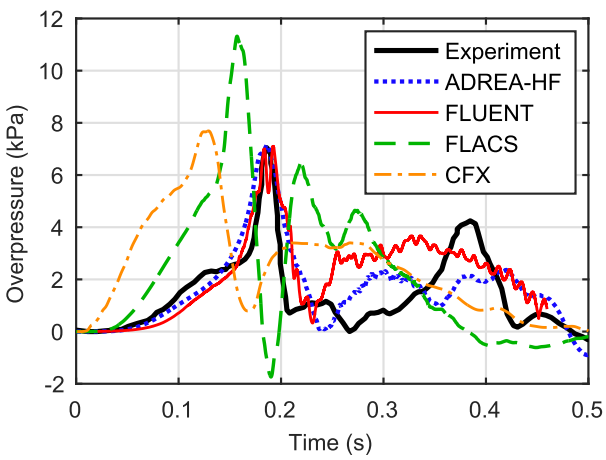

Fig. 10. Overpressure time series inside the enclosure.

overpredicts it by approximately 60%. Moreover, the arrival time of the peak occurs earlier than the experiment. In CFX simulation, the model predicts the magnitude of the peak fairly accurately, but it significantly under predicts the time at which the peak occurs. Moreover, we observe that the pressure increases too early compared to the experiment leading to a broad signal. The reason for that is the fact that the instability driven flame acceleration over the early stage of the deflagration is not explicitly modelled. Instead, the initial flame acceleration is approximated by assuming an artificially high level of turbulence (Section 3.2). Given the lack of physically sound model for the initial flame acceleration, it was not expected that the pressure signal be accurately reproduced. It is therefore very surprising that such a model manages to predict the peak overpressure with good accuracy.

The predicted pressure traces after the first peak are qualitative similar in ADREA-HF, FLUENT and CFX simulations. The pressure decreases to near atmospheric level and then gradually increases to small values (about 2 kPa in the ADREA-HF case and 3 kPa in FLUENT and CFX cases) that are preserved for some period of time. On the other hand, in the FLACS simulation a second pressure peak appears when the flame exits the end of the porch area outside the chamber vent, followed by a smaller third peak.

In Table 4 the maximum overpressure predictions of the CFD models along with the predictions of empirical correlations and the NFPA 68 standard are presented. The predicted-to-experimental pressure ratio is also shown. We observe that the empirical correlations of Bauwens et al. (2012) and Molkov and Bragin (2015) (authors' estimations) give better predictions than the NFPA 68 standard. Their accuracy is comparable to the CFD models. However, we should note that the current experiment was used to calibrate both of these empirical correlations. The Bauwens et al. (2012) correlation takes into account the ignition position and predicts the overpressure peaks generated by the external explosion and by flame-acoustic interactions (in Table 4 the overpressure peaks generated by the external explosion are presented). On the other hand, the correlation by Molkov and Bragin (2015) predicts a single overpressure peak and does not take into account the ignition position. Molkov and Bragin (2015) also provide a conservative estimate of the maximum overpressure (conservative across the complete range of experiments used to fit their correlation), which is also presented in Table 4. The conservative maximum overpressure estimation is 2.6 times higher than the non-conservative prediction. Details about the empirical correlations are presented by Bauwens et al. (2012), Molkov and Bragin (2015) and references within. Finally, the prediction of the NFPA 68 vent sizing standard (NFPA 68, 2013) is presented in Table 4. We observe that the standard is very conservative. The maximum overpressure is 8 times the experimental one. This model was developed for vent design and as a result is expected to be conservative for safety reasons.

Fig. 11 presents the flame speed as a function of distance from the ignition point. In the experiment, there is an initial acceleration phase in the first 2.5 m (just beyond the vent at 2.3 m), followed by a relatively steady period until 5 m away from the ignition location, during which time the flame speed is about 62 m/s, followed by a final sharp increase in the flame speed. We observe that in general the flame speed predictions are similar to the experimental measurements. ADREA-HF,

Table 4
Maximum overpressure predictions (in accessing order for each type of model).

Type of model	Code/Model	Maximum overpressure (kPa)	Predicted to experimental ratio
CFD	FLUENT	7.10	1.01
	ADREA-HF	7.15	1.02
	CFX	7.69	1.10
	FLACS	11.3	1.61
Empirical correlations	Bauwens et al. (2012)	7.83	1.12
	Molkov and Bragin (2015)	8.19	1.17
	Molkov and Bragin (2015) conservative	21.3	3.04
Vent sizing standard	NFPA 68	56.7	8.10

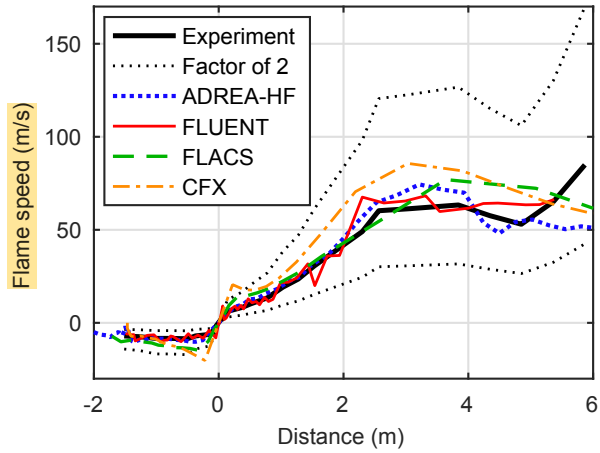

Fig. 11. Flame speed as a function of distance from the ignition point.

FLUENT and FLACS simulations achieve the best agreement with the experiment from 0 to 2.5 m. However, some oscillations are observed in the FLUENT predictions which are thought to be a consequence of post-processing of simulation results obtained using tetrahedral grid. The CFX model over predicts the velocity during this time while all four models over predict the flame speed during the steady period. FLACS and CFX produce the largest over prediction during the steady period. An over prediction of the velocity (in absolute value) is also observed in the FLACS and CFX simulations for the flame front in the direction of the back wall of the enclosure (negative velocities in the figure). The ADREA-HF and FLUENT predictions achieve the best agreement with the experiment in this region and in the non-accelerating region from 2.5 to 5 m. None of the models predicts the final sharp increase in the flame speed seen after 5 m.

In overall, ADREA-HF and FLUENT seems to be in better agreement with the experiment than CFX and FLACS, especially in pressure predictions. ADREA-HF and FLUENT use the similar combustion model along with the LES approach for turbulence, whereas, CFX and FLACS use different combustion models and the RANS approach (k-ε model) for turbulence (Table 3). The reason for the better agreement with the experiment may be attributed to the combustion model that is used and not to the LES approach. This can be supported by the deflagrations simulation performed by Tolias et al. (2014, 2016), where both the LES and the RANS methods were tested (along with the same combustion model that is used in this work) and no significant differences between the results were found.

Fig. 12 shows the model predictions of the pressure inside the enclosure and outside the enclosure at four distances from it. The Figure shows that for all of the models the predicted peak internal pressure occurs shortly after the time at which the flame front is predicted to have exited the vent (shown by the dashed vertical line). This point in time corresponds to the initiation of the external explosion, which is consistent with the theory that the external explosion of the unburned combustible mixture outside the vent contributes to the rise of the pressure inside the enclosure. Particularly pronounced pressure oscillations in FLUENT simulations are believed to be result of RT instability mechanism implemented in the used combustion model: as pressure waves pass the numerical flame front causing its acceleration and deceleration, the RT instability model responses in combustion augmentation and relaxation, which amplifies pressure oscillations.

A view of the flame front (hydrogen volume fraction) when the flame front reaches the vent and when the external explosion occurs is

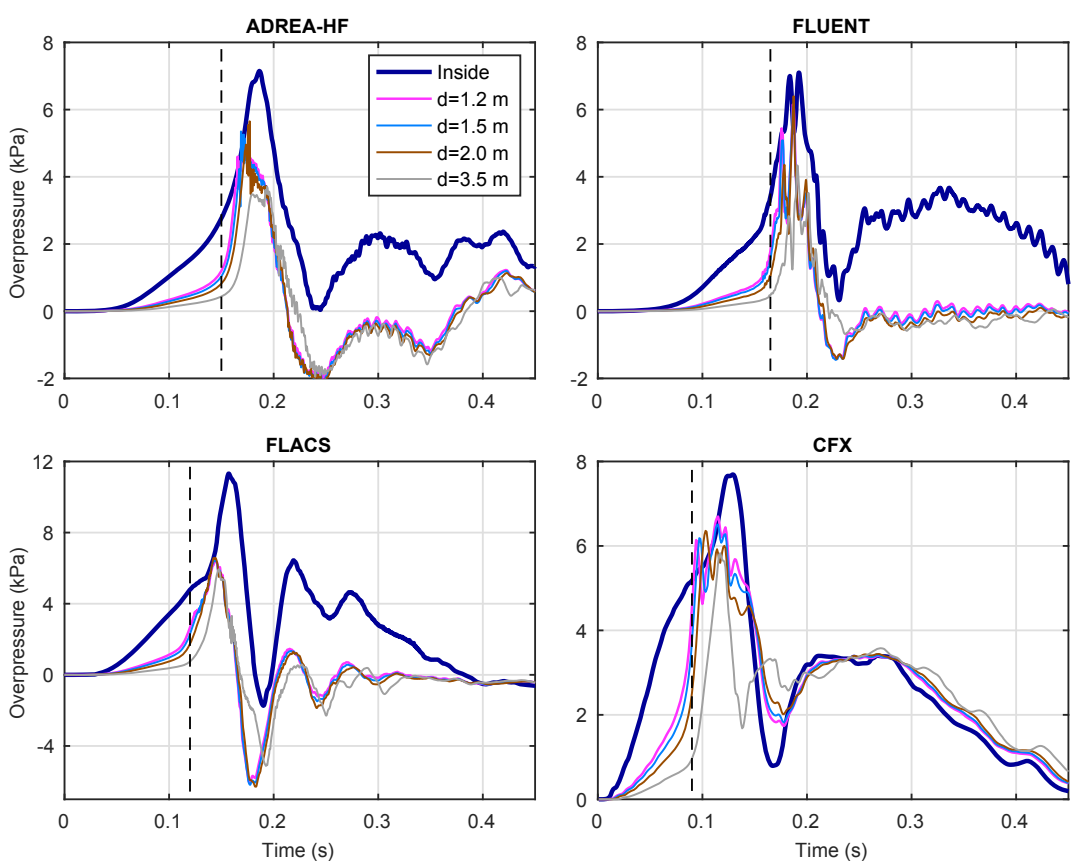

Fig. 12. Predicted internal enclosure pressure and external pressure time series at 1.2, 1.5, 2.0 and 3.5 m from the vent. The vertical dashed line indicates the time at which the flame front is predicted to have reached the vent, which corresponds to the onset of the external explosion.

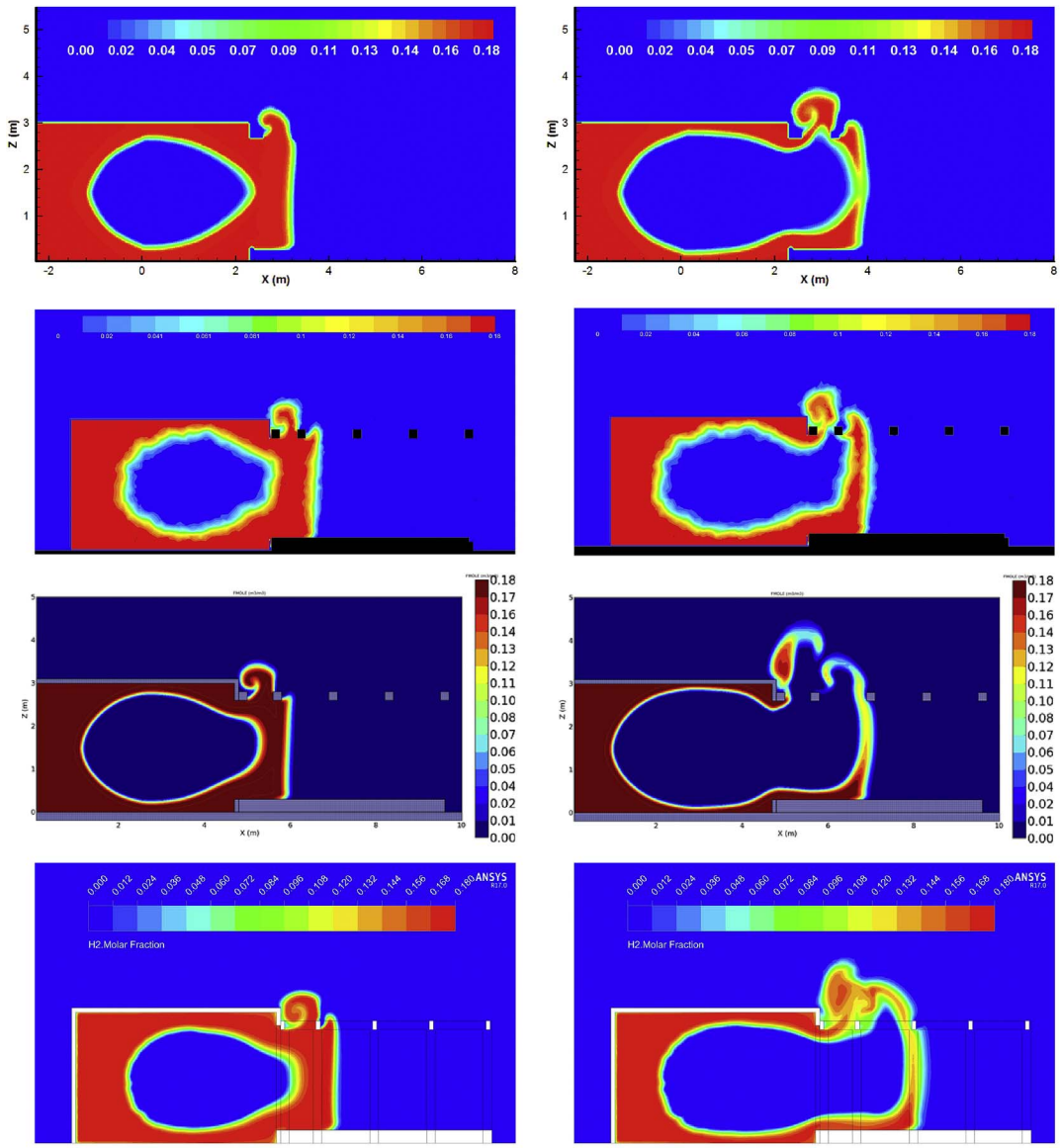

Fig. 13. Hydrogen volume fraction contours in xz plane passing through the ignition point from ADREA-HF, FLUENT, FLACS and CFX simulations (top to bottom) at the time (0.150, 0.165, 0.120 and 0.090 s) when the flame front reaches the vent (left) and at the time (0.170, 0.179, 0.140, 0.120 s) when the external explosion occurs (right).

presented in Fig. 13. An oval-like shape of the flame is formed because of the presence of the vent. We observe that the flame has a similar predicted shape in all models. The unburned mixture forced through the vent is clearly evident at the time at which the flame front reaches the vent. We observe that there is minimal dilution of this mixture with the surrounding air.

Fig. 14 presents the fluctuating velocity, u', time series at several distances from the ignition point for all simulations. Moving away from the ignition point, the peak value at each sensor (in general) increases due to increased turbulence and the peak occurs at a later time due to the propagation of the flame front. Comparing the values among the different simulations shows significant discrepancies between the different models used. This can be attributed to the different turbulence and combustion models.

Fig. 15 presents the fluctuating velocity, u', contours at the time when the flame front reaches the vent and at the time when the external explosion occurs. The ADREA-HF and FLUENT simulations predict similar contours with the higher values observed at the flame location. The shape of the FLACS and CFX predicted contours is significantly different to those obtained using FLUENT and ADREA-HF. The different turbulence approach (k-ε model and LES) may be responsible for this behavior. Differences in u' values between LES and k-ε turbulence models have been observed in other deflagration simulation studies (Tolias et al., 2014, 2016), where the comparison was made using the same combustion model.

The turbulent burning velocity is the main variable driving the flame propagation and rate of combustion and is thus the most important parameter in the simulations. However, diagrams with burning velocity are rarely provided in similar studies. Fig. 16 presents the turbulent burning velocity at several distances from the ignition point for the ADREA-HF and FLUENT simulations. The peak turbulent burning velocity at a given distance from ignition occurs as the flame front passes through that point. This coincides with the period during which combustion occurs. The differences observed between the simulations before and after that period have no practical importance because the combustion rate is almost zero due to the very low value of the progress variable gradient. The discrepancies that are observed in the peak values can be attributed to the different modeling approach used to account for the influence of the Rayleigh-Taylor instability.

Fig. 17 presents the flow velocity magnitude contours, at the time

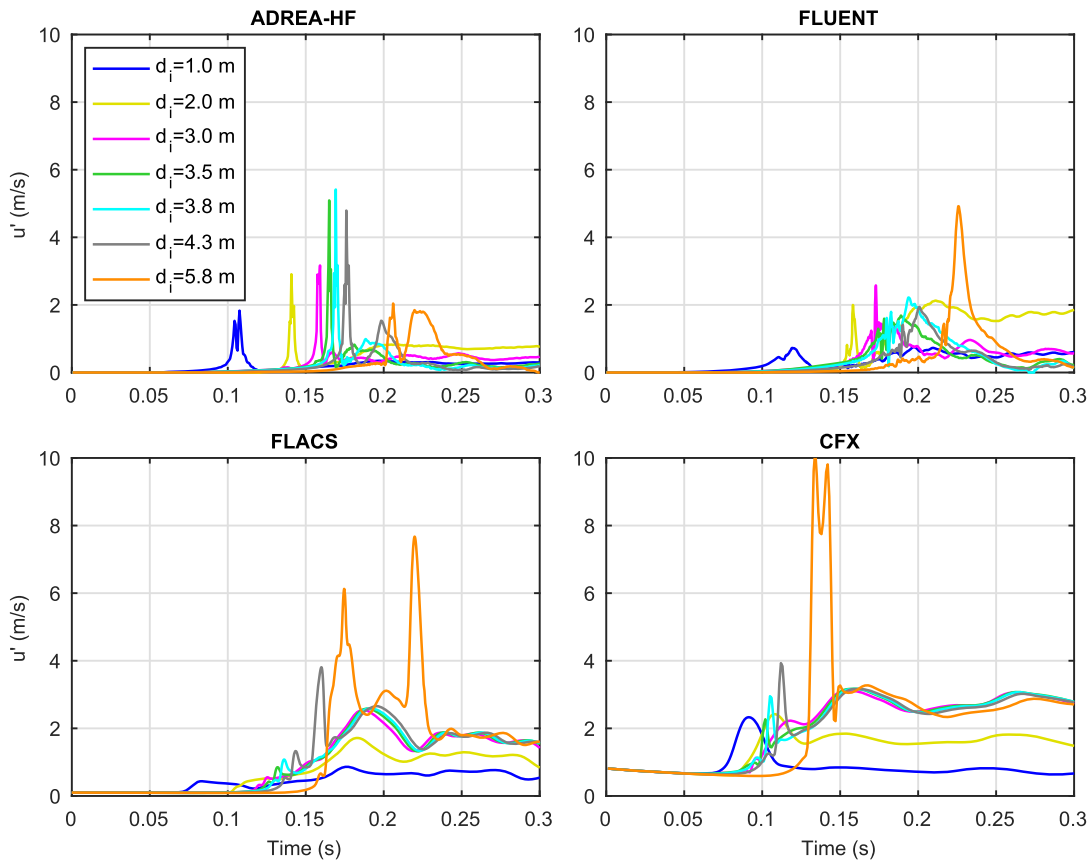

Fig. 14. Time series of turbulent fluctuating velocity u' at 1.0, 2.0, 3.0, 3.5, 3.8, 4.3 and 5.8 m from the ignition point.

when the flame front reaches the vent and at the time when the external explosion occurs. The shape of the contours is similar for all of the models used, although there are differences in the predicted velocity magnitude. High values are observed around and outside of the vent due to flame acceleration and the external explosion. The velocities at the time when the external explosion occurs are higher than at the time when the flame reaches the vent, as expected. The largest velocities are predicted in the FLACS and CFX simulations and then by FLUENT and ADREA-HF.

5. Conclusions

Four different CFD modeling approaches were used to simulate a hydrogen-air deflagration experiment in a 64 m^3 vented enclosure. The case with central ignition, 18% H_2/air mixture and large vent area was studied. In the experiment, two overpressure peaks were observed: the first one corresponds to the external explosion and the second one to the flame/acoustic interaction and structural response. This study focused on the prediction of the first overpressure peak that is generated by external explosion. This peak was the highest one and thus it is more important to be accurately predicted. The second peak (which can be the dominant one in other cases, e.g. when the ignition takes place near the vent) could not be reproduced by the models, as expected.

Sensitivity analyses of several parameters were conducted in order to examine their impact on the results. In FLACS, the final results were found to be reasonably independent of grid resolution. CFX showed some sensitivity to the grid resolution in the overpressure predictions, although the corresponding flame speed predictions were less sensitive. ADREA-HF revealed the smallest dependence of the results on the grid size. Sensitivity analysis on the turbulence initial conditions was also conducted in FLACS and CFX simulations. In FLACS, the results were shown to be sensitive to the initial turbulence length scale. It is anticipated that the initial turbulence parameters could be tweaked to obtain a result which more closely replicates the measured data, although to achieve this would mean using settings which fall outside the range of suggested values. In CFX, a large level of sensitivity on the initial turbulence conditions was found. This was expected because artificially high levels of initial turbulent kinetic energy and turbulent eddy dissipation were specified due to the unsuitable of the combustion model to model low turbulence conditions (which are presented during the early stages in the examined case). Whilst the results appear encouraging, further work would be required before this approach could be used for practical safety assessment applications.

The most significant variable relating to safety applications, the maximum overpressure, is predicted fairly accurately in simulations using ADREA-HF, FLUENT and CFX, while it is conservatively predicted by FLACS. The comparison of the empirical correlations revealed that the models of Molkov and Bragin (2015) and Bauwens et al. (2012) achieve very good agreement with the experiment. However, it should be emphasized that the experiment used for this study has been used in the calibration of both of these models. The peak overpressure prediction made using the NFPA 68 vent sizing standard is very conservative, exceeding the measured pressure peak by more than a factor of eight.

All of the CFD models considered appear to predict the effect of the external explosion. The high intensity of the external explosion caused an increase of the pressure at the area outside the vent which led to the increase of the pressure inside the enclosure. Regarding the flame speed time series, good qualitative agreement between the predictions and measurements were obtained for all models.

In conclusion, this work has shown that CFD models have the potential to capture explosion phenomenon with reasonable accuracy. However, more extensive model validation incorporating further sensitivity studies is required before confidence can be placed in the reliability of CFD models for use in explosion safety assessments. Further

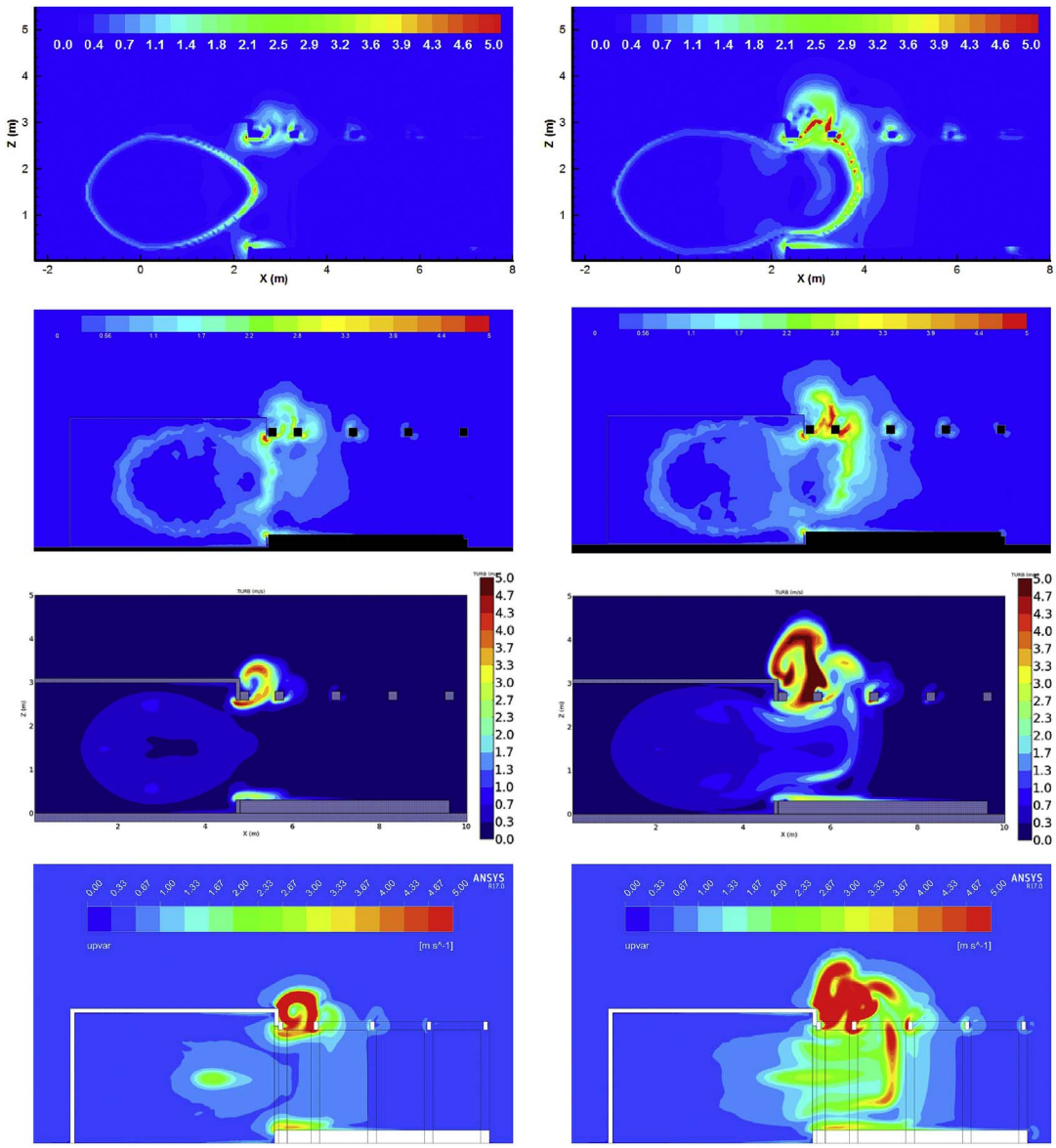

Fig. 15. Turbulent fluctuating velocity u' (from 0 to 5 m/s) contours in xz plane passing through the ignition point from ADREA-HF, FLUENT, FLACS and CFX simulations (top to bottom) at the time when the flame front reaches the vent (left) and at the time when the external explosion occurs (right).

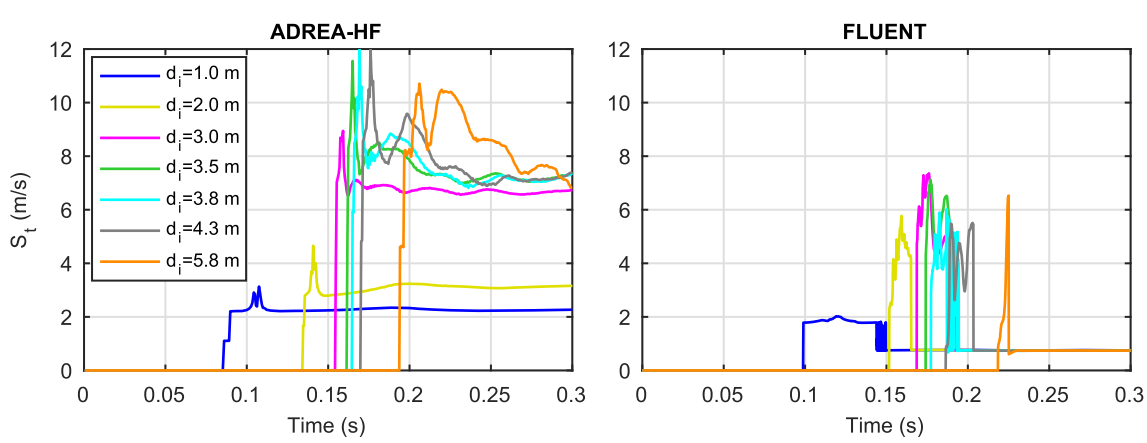

Fig. 16. Turbulent burning velocity S_t at 1.0, 2.0, 3.0, 3.5, 3.8, 4.3 and 5.8 m from the ignition point.

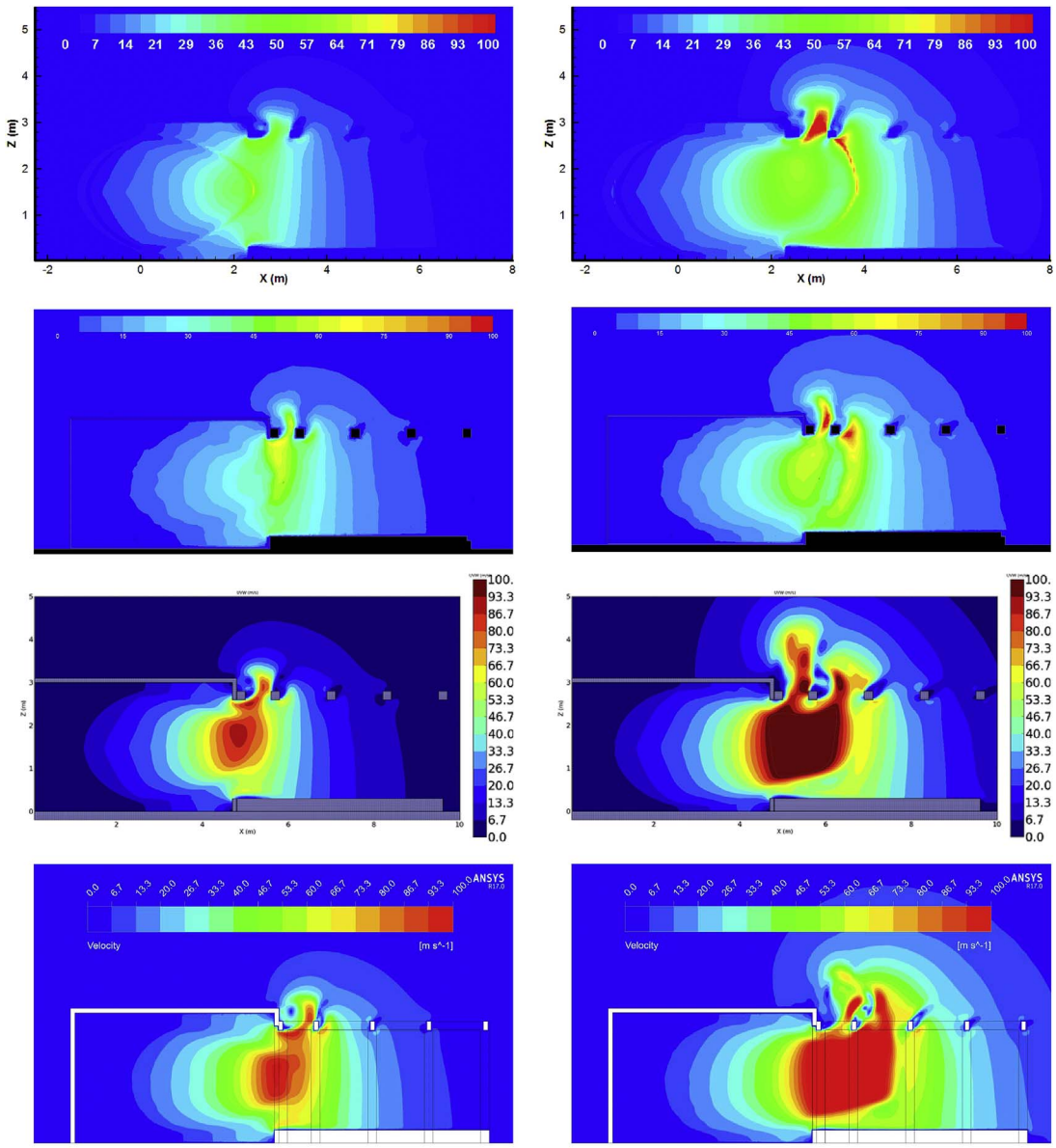

Fig. 17. Velocity contours (from 0 to 100 m/s) in xz plane passing through the ignition point from ADREA-HF, FLUENT, FLACS and CFX simulations (top to bottom) at the time when the flame front reaches the vent (left) and at the time when the external explosion occurs (right).

validation focusing on scenarios of practical interest and model benchmarking exercises ideally involving blind prediction studies are needed to fully evaluate the available explosion models.

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